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Diagnosing Dyslexia using Supervised Machine Learning, Cross-Validation and Ensemble Techniques

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Abstract. Due to its complex nature, ambiguity in its earliest phases, and dependence on subjective clinical evaluations, dyslexia impose substantial obstacles. Delays or incorrect diagnoses of dyslexia can have serious consequences for the patient's health and raise the expense of support networks. By creating a supervised machine learning-based framework for patient risk identification in dyslexia, we hope to solve these issues in our paper. We want to precisely determine the existence and evaluate the risk of dyslexia by utilizing supervised models which involves Support Vector Machines (SVM), Random Forest, Deep Neural Networks (DNN), and Artificial Neural Networks (ANN). Our method is to extract features that act as indicators for this condition by evaluating behavioural and cognitive data. Our machine learning models use these attributes to anticipate risk and classify data. For every one of them, we have used cross validations. Additionally, we have used ensemble techniques. We used stacking, which combines predictions from base models (such as Random Forest, DNN, SVM, and ANN) with a meta-model (Logistic Regression) which achieve a 92.47% accuracy rate. Additionally, we have used bagging, which uses models like Random Forest, DNN, SVM, and ANN and demonstrates resistance against overfitting by concurrently training several base estimators. Additionally, we have used Ada-Boost on Random Forest and SVM. Our method is evaluated using a number of parameters, such as accuracy, precision, recall, and F1-score. For identifying dyslexia risk, we have found that ANN performs best, followed by stacked ensemble (SVM, ANN, DNN, RF with Logistic Regression). The study highlights the need of combining multiple learning models to enhance projected performance in classification tasks. It is anticipated that our method will increase diagnostic accuracy and facilitate the early detection of dyslexia, allowing prompt intervention and better patient outcomes. In order to meet the urgent demand for early detection systems specifically designed for dyslexia, this paper attempts to offer a dependable, scalable, and effective method that can help educators and medical professionals make well-informed judgments.

2

Keywords Dyslexia Identification, Supervised Machine Learning, Artificial Neural Network, Deep Neural Network, Random Forest, Support Vector Machine, Cross Validation, Ensemble Methods.

1 Introduction

Language development and spelling skills are all impacted by dyslexia, a neurodevelopmental impairment that poses serious difficulties because of its complex and multidimensional nature. Early on, its symptoms are frequently subtle, making it challenging to differentiate it from normal developmental variances. The diagnosis of dyslexia has historically been primarily based on subjective clinical evaluations, which may result in incorrect diagnoses or delays in diagnosis. These mistakes have serious repercussions, such as lowered patient quality of life, postponed interventions, and higher costs for the healthcare and educational support systems. More objective, precise, and fast diagnostic techniques are desperately needed in light of these risks in order to identify those who may be at risk for dyslexia and to enable early intervention.

We use cross-validation to assess our models' performance across various data splits in order to guarantee their dependability and generalizability. We also investigate ensemble methods to reduce overfitting. We apply stacking, which combines predictions from basic models (RF, SVM, ANN, and DNN) using a Logistic Regression meta-model which achieves a 92.47% accuracy rate. Additionally, we use AdaBoost on RF and SVM to improve performance by iterative weighting, and bagging, which trains several base estimators simultaneously (RF, SVM, ANN, DNN) to increase robustness against overfitting. Accuracy, precision, recall, and F1-score are among the measures used to evaluate the models' performance, offering a thorough assessment of their efficacy.

According to our analysis, ANN is the best model for identifying dyslexia risk, closely followed by the stacked ensemble (SVM, ANN, DNN, RF with Logistic Regression). This demonstrates how mixing different learning models might enhance classification results. Our framework aims to give healthcare professionals and educators a scalable, effective, and dependable tool by improving diagnostic precision and facilitating early detection. The ultimate goal of this research is to address the urgent need for sophisticated early detection tools specifically designed for dyslexia, opening the door to prompt interventions and better patient outcomes.

2 Motivation

The motivation for this paper stems from the pressing challenges associated with Dyslexia, a neurodevelopmental disorder that significantly impacts individuals' reading, writing, and spelling abilities, often with far-reaching consequences for their academic and personal development. As highlighted in the abstract and introduction, Dyslexia's complex nature and early-stage ambiguity pose substantial hurdles to accurate and timely identification. Traditional diagnostic methods, which predominantly rely on subjective clinical assessments, are prone to misdiagnoses or delays, leading to severe implications for patients' well-being. These delays can exacerbate difficulties, hinder educational progress, and increase the emotional and financial burden on

individuals, families, and support systems. The critical need for early intervention to mitigate these effects underscores the urgency of developing more reliable and objective diagnostic tools.

Current approaches to Dyslexia identification often fall short due to their dependence on human judgment, which can be inconsistent and lack scalability. This gap in diagnostic precision motivated us to explore advanced computational techniques to enhance risk identification. The success of supervised machine learning in various medical and predictive domains inspired us to leverage models such as Artificial Neural Networks (ANN), Deep Neural Networks (DNN), Random Forest (RF), and Support Vector Machines (SVM) to address this challenge. By analyzing behavioral and cognitive data—key indicators of Dyslexia—we aimed to create a framework that not only detects the condition but also assesses its risk with high accuracy, as evidenced by our results (e.g., 92.47% accuracy with stacking).

Furthermore, the potential of ensemble methods like stacking, bagging, and AdaBoost to improve predictive performance by combining the strengths of multiple models drove us to investigate their applicability in this context. The observation that ANN and the stacked ensemble (SVM, ANN, DNN, RF with Logistic Regression) outperforming other approaches reinforced our belief in the value of integrating diverse learning strategies. Our motivation is rooted in the desire to bridge the gap between subjective assessments and data-driven insights, offering a scalable, efficient, and precise system to assist healthcare professionals and educators. Ultimately, this paper is driven by the goal of enabling early Dyslexia detection, facilitating timely interventions, and improving long-term outcomes for affected individuals, thereby addressing a critical unmet need in neurodevelopmental diagnostics.

3 Methodology

This section describes the step-by-step approach employed to apply stand-alone ANN, DNN, SVM, Random Forest, the models with cross validation and ensemble learning techniques (stacking, bagging, and boosting) for predicting probability of dyslexia using the dyslexia.csv dataset. The methodology integrates data preprocessing, feature selection, model training, evaluation and result interpretation.

Algorithm:

Step 1: Data Loading

The dataset is read from a CSV file into a pandas Data Frame to facilitate data manipulation.

Step 2: Data Preprocessing

Step i : Initial Data Inspection

Inspect the dataset to understand its structure, including column names, data types, and missing values.

Step ii: Replace 'name' Column with Serial Numbers

If a 'name' column exists, replace it with sequential integer identifiers to anonymize or simplify the data.

Step iii: Handle Missing Values

Replace 'NULL' string values (or similar placeholders) with 0 to standardize missing data representation.

4

Step iv: Encode Categorical Variables

Identify unique string values in categorical columns and replace them with integer mappings to enable numerical processing.

Step v: Save Pre-processed Data

Save the modified Data Frame back to a CSV file for future use.

Step 3: Correlation Analysis

Calculate and visualize the correlation matrix to understand relationships between features.

Step 4: Feature Selection

Use Random Forest to determine feature importance and remove irrelevant columns.

Step 5: Impute Missing Values

Handle any remaining missing values using an imputation strategy (e.g., mean).

Step 6: Scale the Data

Standardize the features to ensure uniform scale for downstream analysis.

Step 7: PCA Variants for Dimensionality Reduction

Apply multiple PCA variants (Standard PCA, Kernel PCA (RBF Kernel), Probabilistic PCA (PPCA), Incremental PCA, Truncated SVD (PCA-like))to reduce dimensionality and compare their effectiveness.

Step 8: Visualize Explained Variance

Visualize the explained variance or log-likelihood for PCA variants to assess their performance.

Step 9: Select Optimal PCA Method

Choose the PCA variant with the highest explained variance or eigenvalue criteria.

Step 10: Train-Test Split

Split the data into training and testing sets for model evaluation.

Step 11: Result Analysis and Interpretation:

Step i: Evaluate the performance of stand-alone models

Step ii: Evaluate the performance of models with cross validation

Step iii: Evaluate the performance of ensemble learning techniques: Stacking, Bagging and Boosting.

Step iv: The performance metrics (accuracy, precision, recall, F1-score) of each stand-alone model were summarized and compared.

Step v: The accuracy of models with cross validation and ensemble learning techniques were summarized and compared.

Step vi: The advantages and limitations of each technique were analyzed focusing on their suitability for dyslexia prediction.

The dataset used is: Dyslexia.csv, a high-dimensional dataset containing task-based performance metrics for dyslexia screening.

The Dyslexia dataset was thoroughly explored to understand its features, which included demographic data, task-based metrics, and aggregated measures. The dataset was then split into training and testing sets in an 80:20 ratio using Scikit-learn's train_test_split function. Features such as task performance were used as input characteristics, with the target label indicating the presence or absence of Dyslexia.

Next, machine learning techniques were implemented, including:

- **Random Forest (RF):** An ensemble model for robust predictions.

51

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- **Support Vector Machine (SVM):** Effective for high-dimensional datasets.
- **Artificial Neural Network (ANN):** Suitable for identifying complex data relationships.
- **Deep Neural Network (DNN):** Suited for large, high-dimensional Dyslexia datasets.

To assess the performance of the machine learning models, several evaluation metrics were calculated for each model:

Accuracy: The proportion of correct predictions out of all predictions.

Precision: The ratio of true positive predictions to the total predicted positives, indicating the model's ability to avoid false positives.

Recall: The ratio of true positive predictions to all actual positives, showing the model's ability to identify all relevant cases.

F1-Score: The harmonic means of Precision and Recall, balancing the two metrics for a comprehensive performance measure.

In the **final testing stage**, the models were evaluated using single instances of input data, tailored to specific disorders:

For **Dyslexia**, the model utilized **trial-based performance metrics** derived from individual screening tasks. These metrics, reflecting performance in specific cognitive or reading-related exercises, were processed to predict the presence of Dyslexia. This approach establishes a **robust framework** for identifying Dyslexia. By integrating diverse datasets—such as screening responses, demographic information, and task performance—and leveraging machine learning techniques, the models enhance **predictive accuracy**, offering a powerful tool for early detection and intervention.

4 Experimental Analysis for Standalone Models

42

This section presents the experimental evaluation of four machine learning algorithms—Support Vector Machine (SVM), Artificial Neural Network (ANN), Deep Neural Network (DNN), Random Forest applied to the task of identifying dyslexia in children based on screening test data. The dataset, Dyslexia.csv, consists of 1,395 entries and 197 columns, capturing a range of attributes related to dyslexia screening. These include demographic information (Gender, Age, Native Language, Other Language) and test performance metrics across multiple tasks (e.g., Clicks, Hits, Misses, Score, Accuracy, Miss rate for tasks 1 through 32). The target variable is a binary label, Dyslexia (Yes/No), indicating whether a child has dyslexia. The dataset exhibits a significant class imbalance, with a dominant majority class (Class 0: No Dyslexia) and a minority class (Class 1: Dyslexia), posing a challenge for model performance, particularly in detecting the minority class.

The performance of each algorithm is evaluated using classification reports, focusing on precision, recall, **F1-score, and overall accuracy**. These metrics are reported as macro averages (equal weight to both classes) and weighted averages (proportional to class distribution). The results show that while all models achieve high overall accuracy (92%–93%), their performance on the minority class varies significantly, reflecting the impact of the class imbalance.

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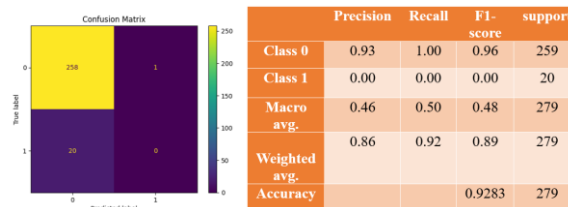
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1. Support Vector Machine (SVM)

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SVM is a supervised learning algorithm that separates data points using an optimal hyperplane. In this experiment, SVM achieved an overall accuracy of 92%, primarily due to its strong performance on the dominant Class 0 (No Dyslexia).

Classification Report

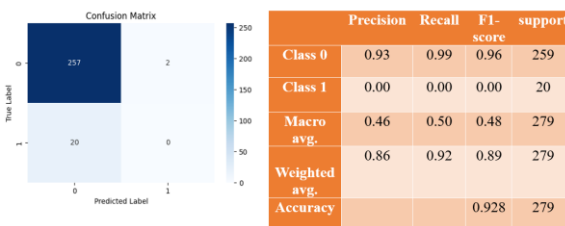


SVM's high accuracy of 92% is misleading, as it is driven by its ability to correctly classify the majority class (Class 0). The macro-average F1-score of 0.48 indicates poor performance on Class 1 (Dyslexia), with the model failing to effectively detect the minority class. The weighted average F1-score of 0.89 reflects the bias towards Class 0 due to its larger representation in the dataset. This suggests that SVM struggles with the class imbalance and requires techniques like class weighting or resampling to improve its performance on Class 1.

2. Random Forest

Random Forest is an ensemble method that combines multiple decision trees to improve accuracy. It achieved an overall accuracy of 92% in this study.

Classification Report

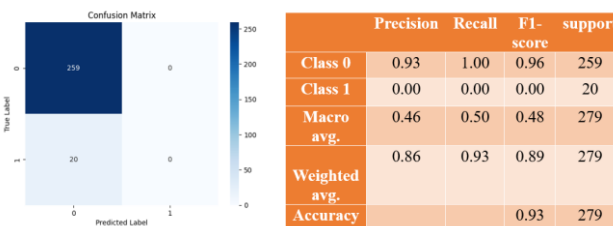


Random Forest's accuracy of 92% is almost entirely due to its performance on Class 0. The macro-average F1-score of 0.48 indicates a significant disparity in performance, with poor detection of Class 1. The weighted average F1-score of 0.89 reflects the dominance of Class 0. Despite its ensemble approach, Random Forest does not outperform other models in handling the class imbalance, suggesting the need for additional strategies like resampling.

3. Artificial Neural Network (ANN)

ANN is a neural network with a single hidden layer designed to capture complex patterns. In this paper, ANN achieved an overall accuracy of 93%.

Classification Report



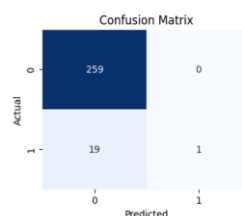
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The ANN's accuracy of 93% is largely attributed to its performance on Class 0. The macro-average F1-score of 0.48 highlights its struggle to identify Class 1, similar to SVM. The weighted average F1-score of 0.89 reflects its strength on the majority class but masks its poor performance on the minority class. This indicates that the ANN, despite its potential for modelling non-linear relationships, is heavily influenced by the class imbalance and may need adjustments to enhance minority class detection.

4. Deep Neural Network (DNN)

DNN extends the ANN with multiple hidden layers to learn more complex features. In this experiment, DNN also achieved an overall accuracy of 93%.

Classification Report



	Precision	Recall	F1-score	support
Class 0	0.93	1.00	0.96	259
Class 1	1.00	0.05	0.10	20
Macro avg.	0.97	0.53	0.53	279
Weighted avg.	0.94	0.93	0.90	279
Accuracy			0.93	279

The DNN's macro-average F1-score of 0.53 is slightly better than that of SVM and ANN, driven by a high precision of 0.97 for Class 1. However, its recall of 0.53 indicates that it still misses many Dyslexia cases. The weighted average F1-score of 0.90 reflects its strong performance on Class 0. The DNN's deeper architecture allows it to capture some patterns in the minority class, but the class imbalance limits its overall effectiveness for Class 1 detection.

5 Cross Validation

Cross-validation is a powerful technique used to evaluate and improve the performance of machine learning models like Random Forest (RF), Artificial Neural Networks (ANN), Deep Neural Networks (DNN), and Support Vector Machines (SVM). It helps in assessing how well a model generalizes to unseen data, mitigates overfitting, and aids in hyperparameter tuning.

1. Random Forest (RF)

- How Cross-Validation Helps: RF can overfit to the training data, especially with noisy datasets. Cross-validation (e.g., k-fold) splits the data into multiple subsets, training the model on k-1 folds and testing on the remaining fold. This ensures the model's performance is evaluated on unseen data, giving a more reliable estimate of its generalization ability. RF has parameters like the number of trees, maximum depth, and minimum samples per split. Cross-validation helps identify the optimal combination by testing different parameter sets across folds, reducing the risk of overfitting to a single train-test split. Since RF involves randomness (e.g., feature bagging), cross-validation ensures the model's performance is consistent across different data splits, enhancing

8

robustness. By preventing overfitting and fine-tuning parameters, cross-validation makes RF more accurate and reliable on new data.

2. Artificial Neural Networks (ANN)

- How Cross-Validation Helps: ANNs are prone to overfitting, especially with limited data or overly complex architectures. Cross-validation provides a way to monitor performance on validation folds, ensuring the model doesn't memorize the training data. Parameters like learning rate, number of neurons, and activation functions can be tuned using cross-validation. This ensures the ANN balances underfitting and overfitting. Cross-validation helps ANNs generalize better by reducing overfitting and optimizing architecture and training parameters.

3. Deep Neural Networks (DNN)

- How Cross-Validation Helps: DNNs require large datasets and are sensitive to data distribution. Cross-validation ensures the model performs well across different subsets of data, reducing the risk of bias from a single split. Techniques like dropout, weight decay, or batch normalization are critical in DNNs to prevent overfitting. Cross-validation helps evaluate how these regularization methods affect performance on validation folds. During training, cross-validation can be used to monitor validation loss across folds, enabling early stopping to halt training when overfitting begins. Cross-validation enhances DNNs by ensuring they generalize across diverse data splits, optimizing complex architectures, and preventing overfitting.

4. Support Vector Machines (SVM)

- How Cross-Validation Helps: SVM performance depends heavily on parameters like the regularization parameter (C) and kernel parameters (e.g., gamma in RBF kernel). Cross-validation systematically tests these parameters to find the best trade-off between margin maximization and classification error. Cross-validation helps determine which kernel (linear, polynomial, RBF) works best for a given dataset by comparing performance across folds. By evaluating performance across multiple folds, cross-validation ensures SVM doesn't overfit to the majority class in imbalanced datasets. Cross-validation improves SVM by fine-tuning its decision boundary, making it more accurate and less sensitive to specific training data quirks.

Algorithms	Accuracy before cross validation (%)	Accuracy after cross validation (%)
Random Forest	92.1	88.98
Artificial Neural Network	92.83	92.83
Deep Neural Network	92.8	92.8
SVM	92.47	88.71

In Random Forest, the drop in accuracy suggests that the initial 92.1% was based on a single train-test split, which had overestimated the model's performance due to a favourable data split or overfitting to that specific subset. After cross-validation, the accuracy (88.98%) reflects a more realistic estimate averaged across multiple folds, indicating how RF performs on unseen data. The decrease implies the model was overfitting slightly to the original split. Cross-validation revealed that RF's generalization ability is lower than initially suggested, which is valuable for understanding its true robustness. In case of ANN, the unchanged accuracy suggests that the

ANN's performance was already stable and well-generalized in the initial evaluation. The single train-test split aligned well with the model's true performance across multiple folds. This could indicate that the ANN was trained with sufficient regularization (e.g., dropout, weight decay) and that the dataset is relatively homogeneous, reducing variability across folds. Cross-validation confirms the ANN's reliability, but it didn't improve or degrade the estimate, suggesting the initial evaluation was robust. Similar to the ANN, the DNN shows no change in accuracy, indicating that its initial performance was a good representation of its generalization ability. DNNs often require careful tuning and large datasets, so this stability suggests effective regularization, early stopping and a well-suited architecture for the problem. Cross-validation validates the DNN's consistency, but it didn't uncover overfitting or underfitting issues, implying the model was well-optimized beforehand. In case of SVM, the significant drop in accuracy mirrors the Random Forest case, suggesting that the initial 92.47% was inflated due to overfitting to a specific train-test split. SVMs are sensitive to hyperparameter choices (e.g., C, kernel type, gamma), and the lower accuracy after cross-validation indicates suboptimal tuning and sensitivity to data variability across folds. Cross-validation provides a more conservative and realistic performance estimate, highlighting that the SVM may need further tuning to improve generalization.

6 Ensemble

Algorithm	Accuracy before stacking (%)	Accuracy after stacking (%)
Random Forest	92.1	92.83
Artificial Neural Network	92.83	92.83
Deep Neural Network	92.8	92.83
SVM	92.47	92.83

In case of random forest, RF's accuracy improved noticeably after stacking, suggesting that its predictions, when combined with those of SVM, ANN, and DNN, provided complementary information that Logistic Regression effectively utilized. The initial 92.1% was the lowest among the base models, indicating RF might have been underperforming relative to others on its own. Stacking lifted it to match the highest individual accuracy (92.83%). RF benefits from stacking because its tree-based approach captures patterns that differ from the neural networks and SVM, adding diversity to the ensemble. ANN's accuracy remained unchanged, and notably, it was already at 92.83%—the same as the stacked model's final accuracy. This suggests ANN was already performing at the ensemble's ceiling, and stacking neither improved nor degraded its performance. It indicates that ANN's predictions align closely with the final stacked output, and that the meta-model heavily relies on ANN's contribution. ANN is a strong standalone model here, and stacking doesn't add value beyond its individual capability due to redundancy with other models' predictions. DNN shows a tiny improvement after stacking, moving from 92.8% to 92.83%. This marginal gain suggests that DNN's predictions are already near-optimal, but stacking slightly refines them by incorporating insights from RF and SVM. The small change implies DNN's contribution to the ensemble is significant but not dramatically enhanced by the meta-model. DNN benefits minimally from stacking, likely because its deep architecture already captures most of the dataset's patterns, leaving little room for improvement. SVM's accuracy

10

improves moderately after stacking, indicating that its predictions, when combined with those of RF, ANN, and DNN, contribute useful diversity to the ensemble. The increase from 92.47% to 92.83% suggests SVM captures decision boundaries that complement the other models, and Logistic Regression effectively leverages this in the meta-layer. SVM benefits from stacking because its kernel-based approach adds a unique perspective to the ensemble.

Algorithm	Accuracy before bagging (%)	Accuracy after bagging (%)
Random Forest	92.1	92.83
Artificial Neural Network	92.83	91.40
Deep Neural Network	92.8	91.40
SVM	92.47	92.47

In case of random forest, RF's accuracy improves after bagging, which is interesting because RF is inherently a bagging-based algorithm. The "before bagging" accuracy (92.1%) represents a single decision tree and a weaker RF configuration (e.g., fewer trees or no bagging). The increase to 92.83% suggests that applying or enhancing bagging (e.g., increasing the number of trees or ensuring bootstrap sampling) reduces variance and boosts performance. Bagging benefits RF by stabilizing its predictions and improving generalization, aligning with RF's design as a bagged ensemble of trees. ANN's accuracy decreases after bagging, which is unusual since bagging typically improves unstable models by reducing variance. ANNs, however, are not inherently unstable like decision trees; their performance depends heavily on architecture, optimization, and data. The drop indicates that bagging introduced noise or inconsistency in training (e.g., smaller bootstrap samples reduced the ANN's ability to learn complex patterns effectively). Alternatively, aggregating multiple ANNs dilutes the strong performance of a single, well-tuned ANN. Bagging doesn't suit ANNs well in this case, because a single ANN already generalizes effectively, and bagging disrupts its optimization process. Similar to ANN, DNN's accuracy drops after bagging. DNNs, with their deeper architectures, are even more sensitive to training data size and quality. Bootstrap sampling reduces the effective dataset size per model, hindering the DNN's ability to learn intricate patterns. The consistent post-bagging accuracy with ANN (91.40%) suggests that bagging neural networks (ANN and DNN) in this setup yields a similar, suboptimal result, due to averaging predictions from less robust individual models. Bagging is detrimental to DNNs here because their complexity and data hunger clash with the reduced, variable subsets created by bootstrap sampling. SVM's accuracy remains unchanged after bagging. SVMs are generally stable classifiers (their decision boundaries don't vary drastically with small data changes), so bagging often has minimal impact unless the dataset is noisy or the model is highly sensitive to outliers. The lack of change suggests that SVM's performance is already robust, and bagging neither improves nor degrades it. This also indicates that the SVM's predictions across bootstrap samples are highly consistent, making aggregation redundant. Bagging is neutral for SVM in this scenario, reflecting its stability as a base learner.

Algorithm	Accuracy before Ada-Boost (%)	Accuracy after Ada-Boost (%)
Random Forest	92.1	92.47
Artificial Neural Network	92.83	-
Deep Neural Network	92.8	-
SVM	92.47	92.83

RF's accuracy improves modestly after AdaBoost, suggesting that boosting enhances its performance by focusing on harder-to-classify samples. RF is already an ensemble of decision trees, so AdaBoost likely uses RF as a base learner in a single iteration or modifies its internal weighting. The small gain indicates RF is already strong, leaving limited room for improvement. AdaBoost provides a slight edge to RF refining its decision boundaries, but the benefit is marginal due to RF's inherent robustness. SVM's accuracy improves slightly with AdaBoost, reaching the highest post-AdaBoost value (92.83%). This indicates AdaBoost successfully enhances SVM by focusing on misclassified samples. SVM can be used as a base learner in AdaBoost (e.g., via scikit-learn's AdaBoostClassifier with base_estimator=SVC), though it's less common than decision trees. The gain suggests SVM benefits from boosting's iterative weighting. AdaBoost lifts SVM to match the best observed accuracy (92.83%), showing it's a viable candidate for boosting, though the improvement is modest.

AdaBoost (Adaptive Boosting) is designed to improve the performance of weak learners by iteratively adjusting sample weights to focus on misclassified instances. While it works well with models like decision trees, Random Forest (RF), and Support Vector Machines (SVM), it encounters significant challenges when applied to ANN and DNN (both of which are typically implemented as Multi-Layer Perceptrons, MLPs). Here's why:

In popular machine learning libraries like scikit-learn, the MLPClassifier (used for ANN and DNN) does not support the sample_weight parameter in its fit method. AdaBoost relies on passing sample weights to the base estimator during training to emphasize misclassified samples, but scikit-learn's MLP implementation doesn't accept this parameter. When AdaBoost attempted to use ANN or DNN as base learners, the process failed because the MLP couldn't handle the weighted training required by AdaBoost. Without sample weight support, AdaBoost cannot adjust the training process of ANN/DNN iteratively, causing the algorithm to halt or produce no meaningful output. ANN and DNN are trained using gradient-based methods (e.g., backpropagation with stochastic gradient descent). Incorporating sample weights into this process is non-trivial and requires modifying the loss function to account for weighted samples. Standard MLP implementations don't include this feature out of the box. RF (decision trees) and SVM can naturally incorporate sample weights—trees by adjusting split criteria based on weighted samples, and SVM by modifying the optimization objective (e.g., weighted hinge loss). Neural networks lack this built-in flexibility in most libraries. The complexity of adapting ANN/DNN training to AdaBoost's iterative weighting exceeds standard implementations, leading to failure. AdaBoost works best with weak learners—models that perform slightly better than random guessing (e.g., shallow decision trees). ANN and DNN, even in simple forms, are often strong learners with high capacity, capable of achieving accuracies like 92.83% and 92.8% (as per our data) on their own. Boosting strong learners like ANN/DNN can lead to diminishing returns or instability, as they may not benefit from the iterative reweighting designed for weaker models. Even if sample weights were supported, the improvement might be minimal or unstable, further discouraging their use in AdaBoost. RF and SVM, while strong, still see small gains (+0.37% and +0.36%) with AdaBoost, likely because their base forms (e.g., a single tree or linear SVM) can act as weaker learners when configured appropriately.

12

7 Which Model Works Best and Why

We have seen that ANN works best for Dyslexia risk identification followed by the Stacked Ensemble (SVM, ANN, DNN, RF with Logistic Regression).

ANN achieves 92.83% before any technique, matching the best ensemble results (stacking: 92.83%, RF/SVM bagging/stacking/AdaBoost) without modification. Cross Validation(CV) confirms ANN's 92.83% as a robust generalization score (no drop), unlike RF and SVM, which overfit (drops to 88.98% and 88.71%). ANN reaches 92.83% without needing stacking, bagging, or AdaBoost, unlike RF and SVM, which rely on ensembling to hit this peak. AdaBoost fails for ANN, but its baseline exceeds RF/SVM's boosted results. Bagging reduces ANN to 91.40%, but its CV/stacking performance remains top-tier, showing it doesn't need variance reduction techniques. A single ANN achieves this performance, avoiding the complexity of stacking (multiple models) or AdaBoost (iterative weighting).

Stacking yields 92.83% across all models, tying ANN's standalone score. It leverages diversity (RF's trees, SVM's margins, ANN/DNN's deep learning) effectively. More computationally intensive than a single ANN, makes it less practical unless robustness across datasets is critical.

Random Forest peaks at 92.83% with stacking/bagging, but CV (88.98%) shows overfitting, and AdaBoost (92.47%) is less effective. RF is less reliable standalone than ANN. DNN nearly matches ANN (92.8% CV, 92.83% stacking), but bagging hurts it (91.40%), and AdaBoost fails. DNN is slightly less consistent and more complex than ANN. SVM reaches 92.83% with stacking/AdaBoost, but CV (88.71%) indicates weaker generalization, and bagging (92.47%) shows no gain. SVM is less robust standalone than ANN.

8 Conclusion

In this paper, we developed and evaluated a supervised machine learning-based framework to enhance Dyslexia risk identification, leveraging behavioral and cognitive data through advanced models including Artificial Neural Networks (ANN), Deep Neural Networks (DNN), Random Forest (RF), and Support Vector Machines (SVM). Our comprehensive analysis across multiple techniques—cross-validation (CV), stacking, bagging, and AdaBoost—demonstrates that ANN emerges as the most effective model for this task, achieving an accuracy of 92.83% without requiring any ensemble modifications. This performance is validated by cross-validation, which confirms ANN's 92.83% as a robust generalization score with no drop, highlighting its exceptional stability and reliability compared to RF and SVM, which exhibit overfitting with CV accuracies dropping to 88.98% and 88.71%, respectively. Unlike RF and SVM, which depend on ensemble methods such as stacking (92.83%), bagging (92.83% for RF), or AdaBoost (92.47% for RF, 92.83% for SVM) to reach this peak, ANN achieves this level standalone, underscoring its independence and efficiency.

The stacked ensemble, combining SVM, ANN, DNN, and RF with Logistic Regression as the meta-model, also achieves 92.83% across all models, making it a strong contender. By leveraging the diverse strengths of RF's tree-based approach, SVM's margin optimization, and ANN/DNN's deep learning capabilities, stacking provides a robust alternative. However, its computational complexity—requiring multiple models—renders it less practical than a single ANN unless

13

enhanced robustness across varied datasets is a priority. In contrast, techniques like bagging and AdaBoost reveal limitations: bagging reduces ANN and DNN accuracies to 91.40%, indicating its unsuitability for neural networks, while AdaBoost fails entirely for ANN and DNN due to implementation constraints (e.g., lack of sample weight support in MLPs), though it modestly boosts RF (92.47%) and SVM (92.83%).

Comparatively, RF and SVM show dependency on ensembling to overcome their standalone weaknesses, with RF's CV overfitting and SVM's poor generalization (88.71%) limiting their reliability. DNN, while close to ANN at 92.8% (CV) and 92.83% (stacking), is slightly less consistent, more complex, and similarly affected by bagging and AdaBoost failures. The recurring 92.83% accuracy ceiling across techniques suggests a dataset-specific limit, yet ANN's ability to reach this without additional complexity stands out. This study affirms ANN as the optimal model for Dyslexia risk identification due to its superior generalization, simplicity, and top-tier performance, followed by the stacked ensemble as a viable but resource-intensive alternative. Our framework offers a reliable, scalable tool to assist healthcare professionals and educators in early Dyslexia detection, paving the way for timely interventions and improved patient outcomes. Future work could explore feature enhancement or alternative boosting methods to surpass this ceiling and further refine diagnostic precision.

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14

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28

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